

in the underlying markets. We estimate our current capacity at \$6 billion to \$9 billion. But we always add the caveat that if volatilities decline by 50 percent on average across the markets, our capacity would be reduced by a similar amount.

How do you control risk?

The core of the risk management is evaluating the risk of each market based on an exponentially weighted moving average of the daily dollar range per contract. This risk management metric has kept our volatility relatively stable near the target level, even through periods of wide gyrations in the markets. One of the things I'm particularly proud of in terms of risk management is that through the chaos of 2008 and 2009 our volatility remained very near our target level of 12 percent annualized.

I assume then that you were trading a much smaller number of contracts in each market per million dollars in 2008 than you normally do?

Absolutely. As volatility increased, the number of contracts we were trading dropped precipitously.

What other risk management procedures do you use besides adjusting the trading size for changes in the volatility of the underlying markets?

The volatility adjustment has worked extremely well for the entire history of the program. The part of our risk management process that has worked poorly during the past two years is our leverage reduction policy. In its initial formulation, whenever there was an intramonth drawdown of 6 percent from a monthly equity peak, we cut our exposure to 75 percent of normal.² On an 8 percent drawdown, the exposure was cut to 50 percent, and on a 10 percent drawdown, it was cut to 25 percent. Then there were analogous rules for increasing exposure back up as the drawdown was reduced. From 2003 through 2009, the leverage reduction rule very slightly lowered our return/risk, but it made it much easier to sleep because whenever we were in a significant drawdown, our trading size would be smaller. However, in 2010 and 2011, our leverage-reduction policy really hurt us badly because the periods when our models were really on fire occurred when our exposure was lowest. To